Novel Biomarker Analysis using Higher Order Feature Extraction: A case study using COVID-19 reveals differential protein expression patterns from disease onset to recovery

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Motivation

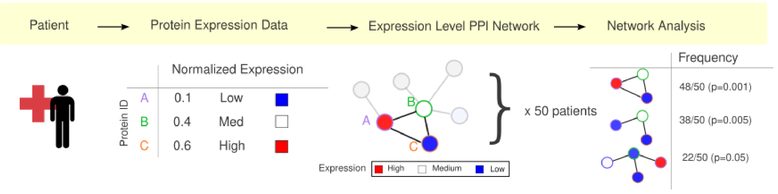
- The majority of proteomic analyses focus on individual proteins.
- Graph networks are a good data modality for expressing complex data, such as protein-protein interactions (PPIs).
- Integrating PPI network graphs with conventional proteomics enriches the dataset by modeling protein expression within the physiological interaction networks in which proteins function.

Data Sources

- Protein-protein interaction (PPI) data from the STRING database[1].
- Plasma protein profiles (1463 proteins) of 50 COVID-19 patients at two timepoints (disease and recovery). Previously published by [2].

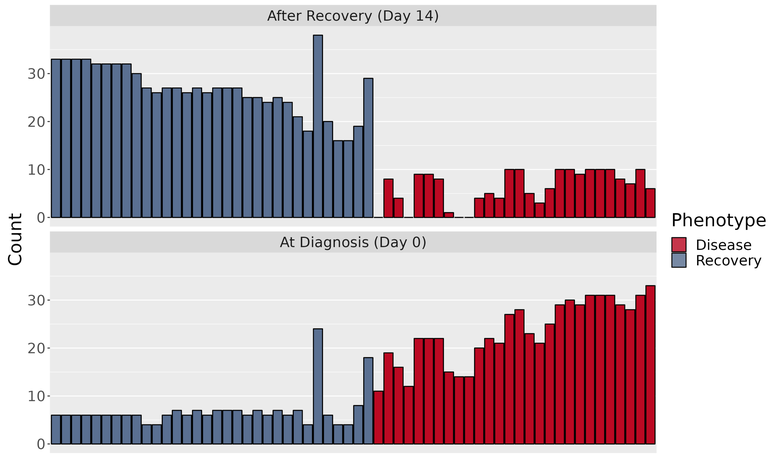
Proteomic Signatures

- **Interaction Graphs:** We construct a graph for each patient sample, representing the state of their proteomic network at each timepoint.
- **Feature Extraction:** We pool the graphs from each timepoint and performed frequent subgraph mining to identify 278 motifs within the data [3].



Pattern Subpopulations Associated With Phenotype

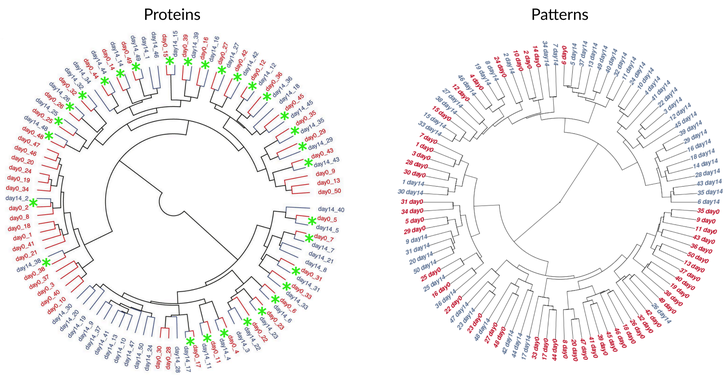
- **Pattern Profile:** We identified protein patterns that are over-represented at each timepoint and construct a motif profile for each sample.



Pattern Profiles Accurately Stratify COVID Status

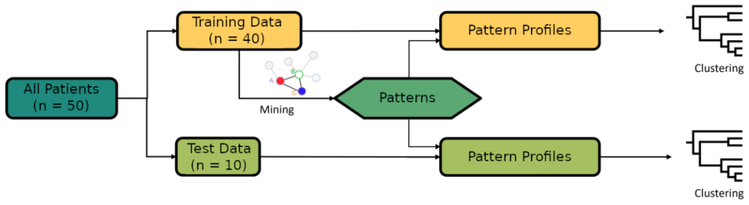
We performed hierarchical clustering using motif profile and protein expression in order assess each feature's ability to classify the patient groups [4].

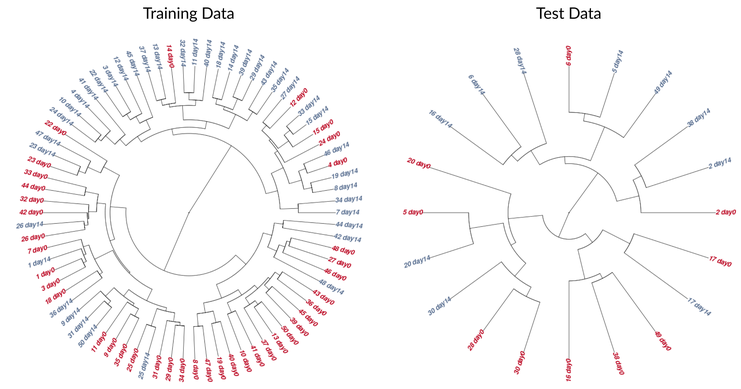
- **Single Protein Profiles:** Majority of patients are most similar to themselves at different timepoints (*).
- **Pattern Profiles:** We find a higher proportion of patients profiles most associated with other patients at the same timepoint.



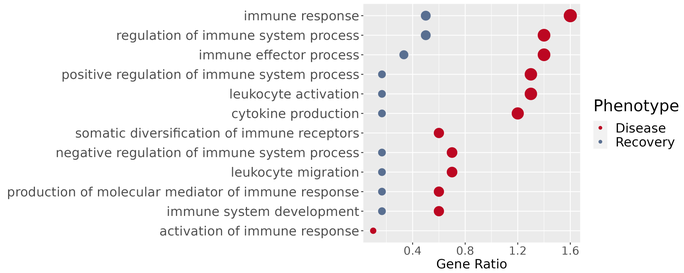
Stratification Remains Accurate on Withheld Data

- A majority of samples are correctly clustered when testing on a withheld subset of samples.





Disease associated Patterns are Enriched with Immune Ontologies



Conclusion and Future Directions

- Protein-expression patterns were created by leveraging PPI networks and plasma protein measurements. This method enriches our understanding of differential protein expression by capturing the physiological interaction networks in which proteins function.
- Further analyses reveal that our methodology enriches for specific proteins like IFNG, TCN2, and ANGPTL1, which are markers for immunomodulatory and innate immune responses [5]. These proteins offer new perspectives on patient responses from the time of infection to recovery.
- Our approach is disease-agnostic. It allows for high-fidelity analysis of differential protein expression across varying timepoints or conditions. By contextualizing proteins within their interaction networks, we can better identify biomarkers indicative of disease response and treatment efficacy.
- By leveraging network analysis, we dissect these PPI landscapes, defining cohesive modules or protein clusters, that adhere more closely to real world outcomes while elucidating affected biological pathways or systems

References

[1] Damian Szklarczyk, Annika L. Gable, Katerina C. Nastou, David Lyon, Rebecca Kirsch, Sampo Pyysalo, Nadezhda T. Doncheva, Marc Legeay, Tao Fang, Peer Bork, Lars J. Jensen, and Christian von Mering. The STRING database in 2021: Customizable protein-protein networks, and functional characterization of user-uploaded gene/measurement sets. Nucleic Acids Research, 49(D1):D605–D612, 2021.

[2] Wen Zhong, Ozlem Altay, Muhammad Arif, Fredrik Edfors, Levent Doganay, Adil Mardinoglu, Mathias Uhlen, and Linn Fagerberg. Next generation plasma proteome profiling of COVID-19 patients with mild to moderate symptoms. EBioMedicine, 74, 2021.

[3] Carlos Oliver, Dexiong Chen, Vincent Mallet, Pericles Philippopoulos, and Karsten Borgwardt. Approximate network motif mining via graph learning. arXiv preprint, 2206.01008, 2022.

[4] Frank Nielsen. Hierarchical Clustering, pages 195–211. 02 2016.

[5] Jamie R Schoenborn and Christopher B Wilson. Regulation of interferon-γ during innate and adaptive immune responses. Advances in immunology, 96:41–101, 2007.